Sajan Kumer Sarker

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Summary

Computer Science and Engineering graduate with internship experience in AI research and fintech, with a growing interest in Artificial Intelligence and Cybersecurity, particularly at their intersection and Code Security. Actively engaged in research exploring Artificial Intelligence, Machine Learning, Large Language Models, and Cybersecurity, with a focus on developing intelligent, secure, and reliable AI-driven solutions. Additionally, gaining industry experience in applying data-driven approaches and AI technologies to real-world challenges in the fintech sector.

Research Interests

- AI for Cybersecurity *(Vulnerability Analysis, Malware Analysis, Intrusion Detection, Threat Detection, Security Automation)*
- Large Language Models *(LLM Reasoning, Model Editing, Retrieval Augmented Generation, Parameter-Efficient Learning)*
- AI Agents *(Multi-Agent Systems, Agent Security, Guardrails, Tool-Augmented LLMs)*

Education

Bachelor of Science in Computer Science and Engineering (BSCSE) February 2021 – September 2025
North South University
– CGPA: 3.04/4.00 (Equivalent to ≈ 84%)
– Thesis: An Explainable Approach for Network Intrusion Detection Using Machine Learning and Deep Neural Network
Poster Report Codes

Professional Experience

N-Risers Intern – Business Assurance Department August 2026 – Present
Nagad, Bangladesh
– Analyze transactional data and build systems for fraud analysis, suspicious activity detection, and business intelligence.
– Develop AI-powered solutions with AI/LLMs, and computer vision to automate workflows and support business operations.

Research Intern *(Part-time)* June 2026 – Present
Elite Research Lab LLC
– Conducting research on secure and trustworthy AI systems, focusing on LLM and AI agent security.
– Contributing to structured reasoning workflows, literature review, experimental design, and technical documentation.

Ongoing Research

1. A Secure Autonomous Agentic Framework for Microservice Code Generation with Automated Vulnerability Detection and Remediation – *Elite Research Lab LLC*

Projects

- CyberSec-Reasoner – A Three-Stage Post-Trained Reasoning LLM for Vulnerability Analysis *GitHub*
  – Developed a post-training pipeline for Qwen3.5-4B integrating Knowledge-Based Supervised Fine-Tuning (SFT), Cold-Start Reasoning SFT, and Group Relative Policy Optimization (GRPO) to improve Chain-of-Thought reasoning and vulnerability-to-weakness (CVE-to-CWE) mapping through reward-guided optimization.

- Multi-lingual YouTube Comment Sentiment & Type Analysis *GitHub*
  – Developed and evaluated a multilingual YouTube comment classification framework using Qwen2.5-14B for automated annotation and XLM-RoBERTa for sentiment and comment-type prediction, achieving 76.68% and 76.05% test accuracy, respectively, with FastAPI-based analytical visualization.

- SPIRICHAT - RAG-Based Q&A Chatbot *GitHub*
  – Implement a RAG-based chatbot integrating Hugging Face embeddings, Pinecone vector search, and OpenAI GPT-3.5 API to retrieve relevant document context and generate context-aware responses through a Flask and LangChain backend.

- Drone Human Detection & Counting System – YOLO26-OBB + BoT-SORT Aerial Analytics Pipeline *GitHub*
  – Developed an end-to-end aerial computer vision pipeline using fine-tuned YOLO26l-OBB and BoT-SORT for human detection, multi-object tracking, and unique person counting in drone imagery, incorporating VisDrone2019-DET preprocessing and oriented bounding box conversion.

- Pre-trained Transformers for Java-Python Code Translation *GitHub*
  – Developed a two-stage Java-to-Python code translation pipeline using fine-tuned CodeT5 and PLBART for code generation, correction, and minimization, achieving a BLEU score of 51.72 and CodeBLEU of 0.4913 with CodeT5.

Skills

Languages	Python, Java, SQL, HTML/CSS.
Libraries/Frameworks	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, FastAPI.
Databases	MySQL, PostgreSQL, Neo4j, PuppyGraph, Vector Databases.
Cloud & Infrastructure	Microsoft Azure, Docker, Virtual Machines, Git, GitHub, Linux, Windows.
Networking & Security	TCP/IP, UDP, HTTP/HTTPS, DNS, SSH, OSI Model, CVE, CWE, Offensive & Defensive Security.
Tools	Weights & Biases (W&B), Postman, LaTeX, Claude Code, Wireshark.
Core Concepts	Object-Oriented Programming, Debugging, Data Structures & Algorithms.
AI/ML & LLMs	Machine Learning, Deep Learning, Large Language Models, Generative AI, Agentic AI, Computer Vision, Graph Neural Networks, Model Quantization, Reinforcement Learning, Prompt Engineering.

Activities

- Google Cloud Gen AI Academy APAC Edition – Hack2Skill × Google Cloud
  – Participated in the Learning Journey, gaining exposure to Generative AI concepts, tools, and practical applications.
  Cohort 1 Cohort 2 Cohort 3
- Aspire Leaders Program – Aspire Institute
  – Completed a global learning experience focused on leadership development, and professional growth. — Credentials
- ML Summer School 2026 – Cohere Labs
  – Attended 12 sessions on modern ML, LLMs, AI agents, and ML research — Credentials

Certifications

- Certified in Cybersecurity (CC) – ISC2 – *(Credentials)* Oct. 2026
- Pre Security Learning Path – TryHackMe – *(Credentials)* May 2026
- Associate AI Engineer for Developers – DataCamp – *(Credentials)* Apr. 2026
- Machine Learning Specialization – Coursera – *(Credentials)* Dec. 2024
- Google IT Automation with Python – Coursera – *(Credentials)* Aug. 2024

Additional Details

- Soft Skills: Problem Solving, Research-oriented, Team Collaboration, Time Management, Self-driven, Presentation Skills.
- Languages: Bengali (native), English (B2), Hindi (A2)

References

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